Liver Transplantation

When to Refer and What's New

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Slides (PDF)

Illustrations generated with Google Gemini



Disclosures

No relevant financial relationships to disclose.

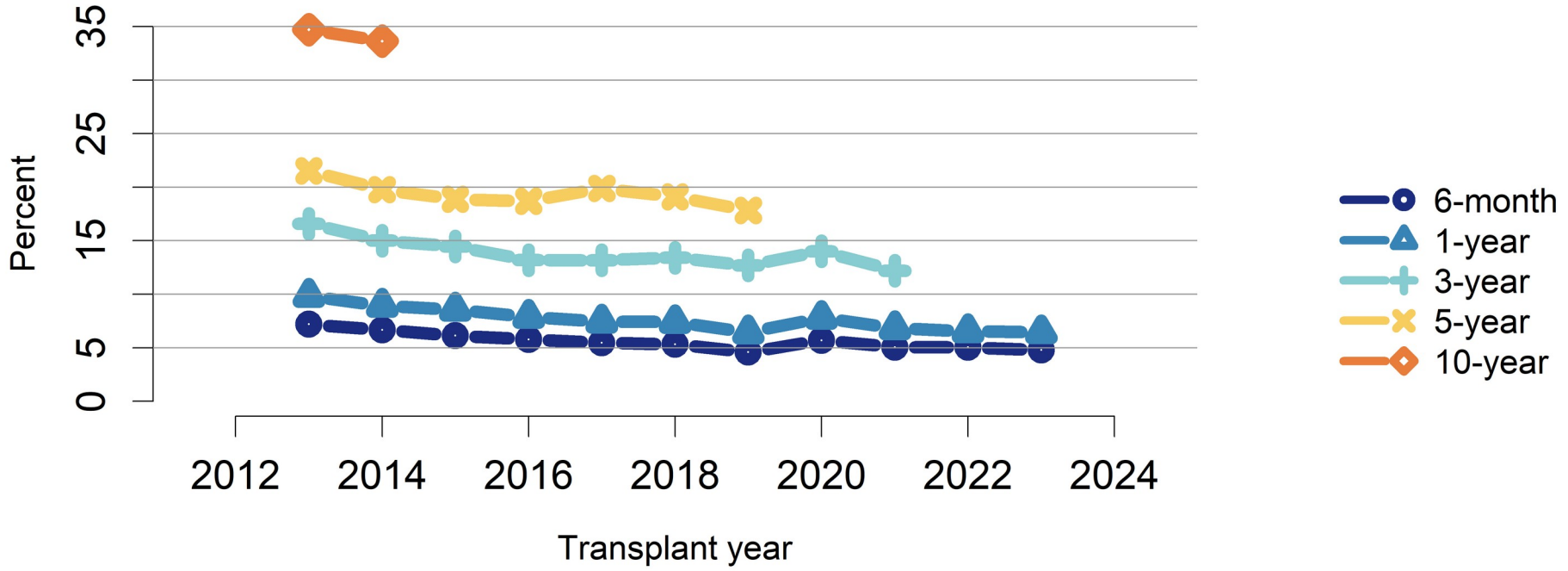
Objectives

- Know when to refer a patient with liver disease for transplant evaluation
- Know who can be a transplant candidate today
- Know where donor livers come from now
- Care for a transplant recipient on your service



Is liver transplant worth all the trouble?

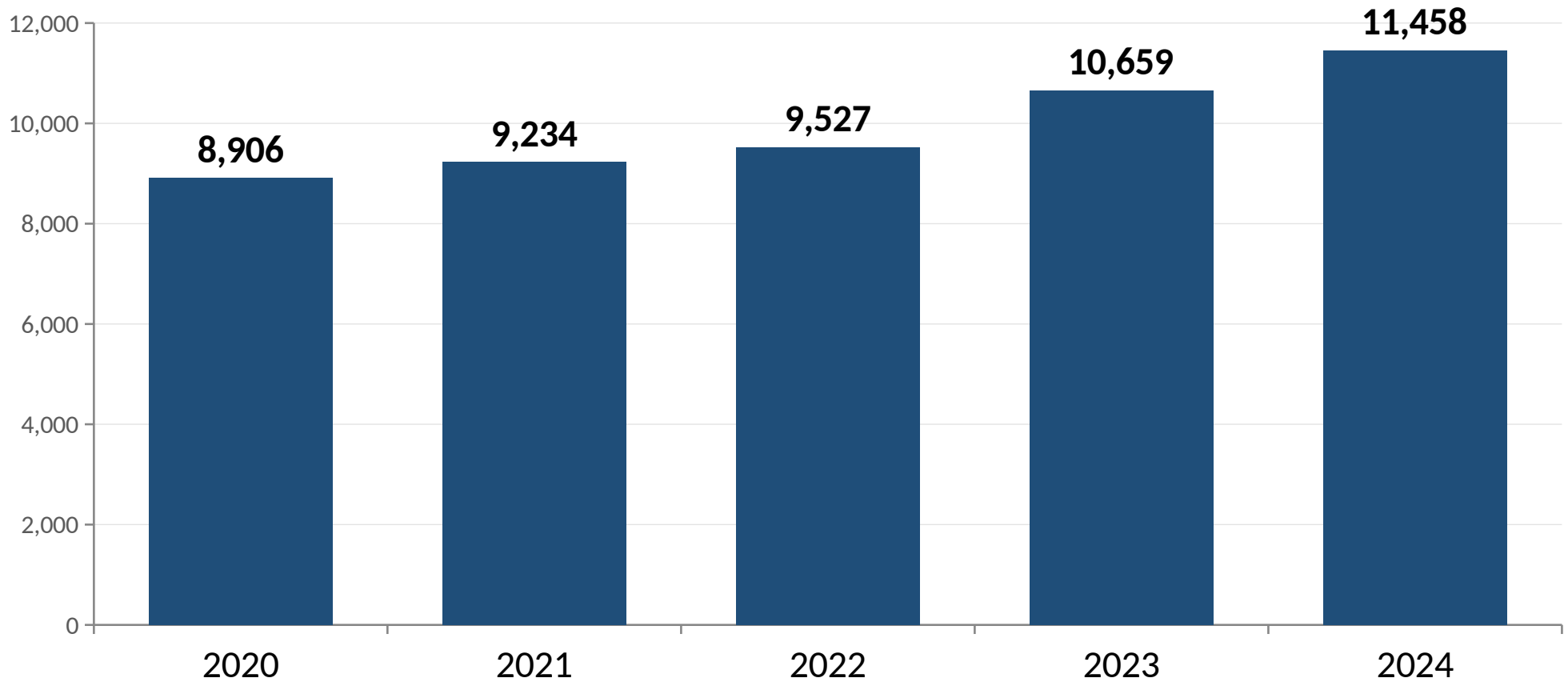
Figure LI 59: Percentages of patient deaths among adult liver transplant recipients



OPTN/SRTR 2024 Annual Data Report

OPTN/SRTR 2024 Annual Data Report, Figure LI 59. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

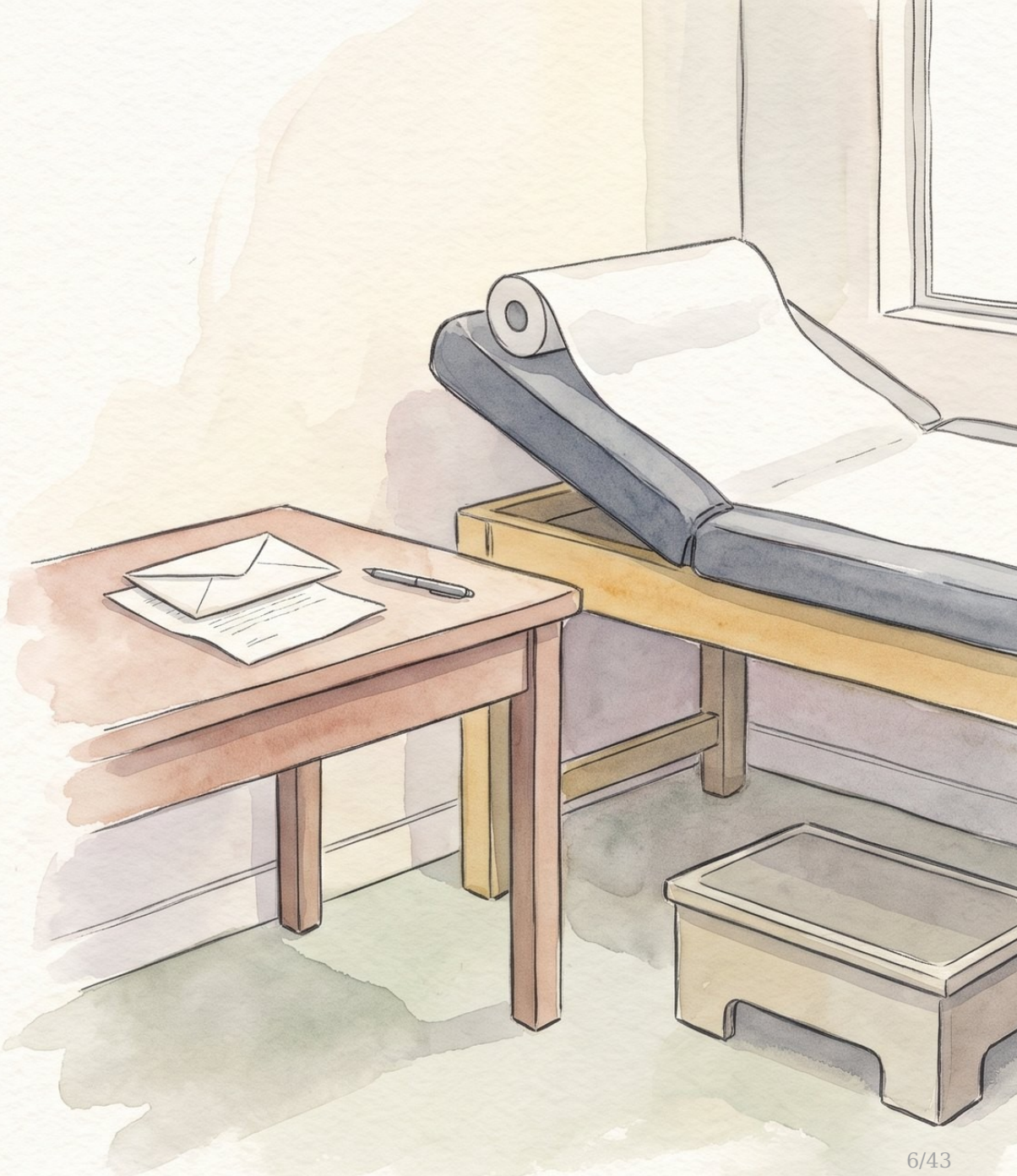
More liver transplants than ever



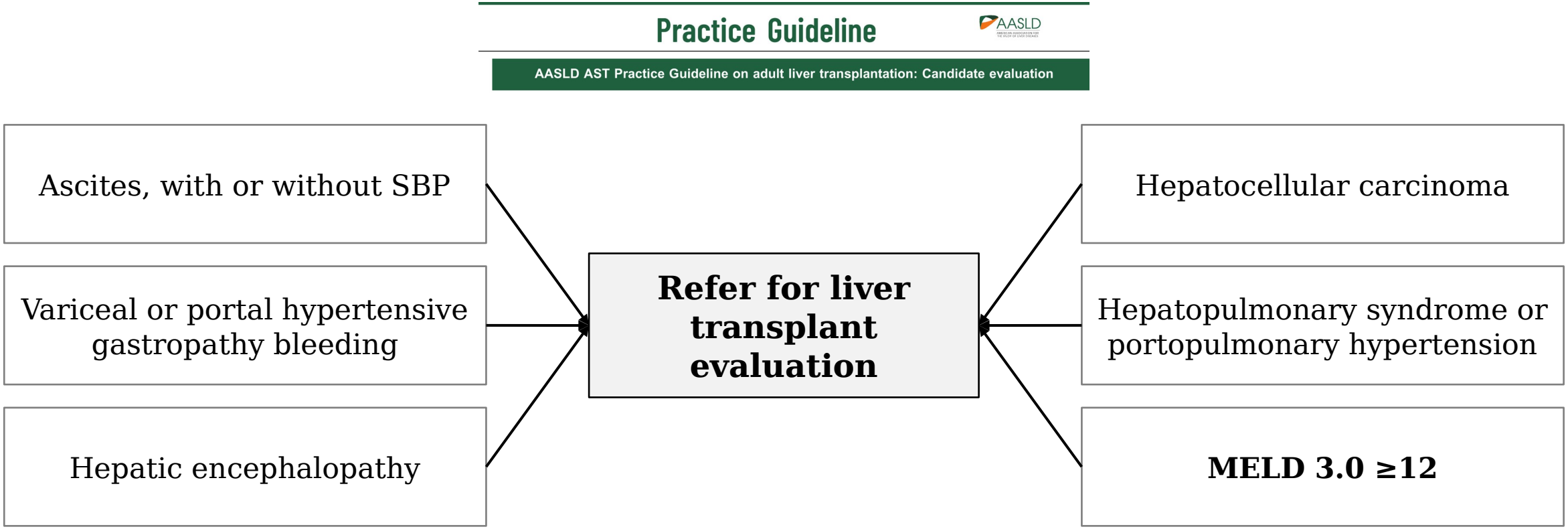
11,458 in 2024 (10,886 adults), 70% more than in 2014

OPTN/SRTR 2020-2024 Annual Data Reports. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

When to refer



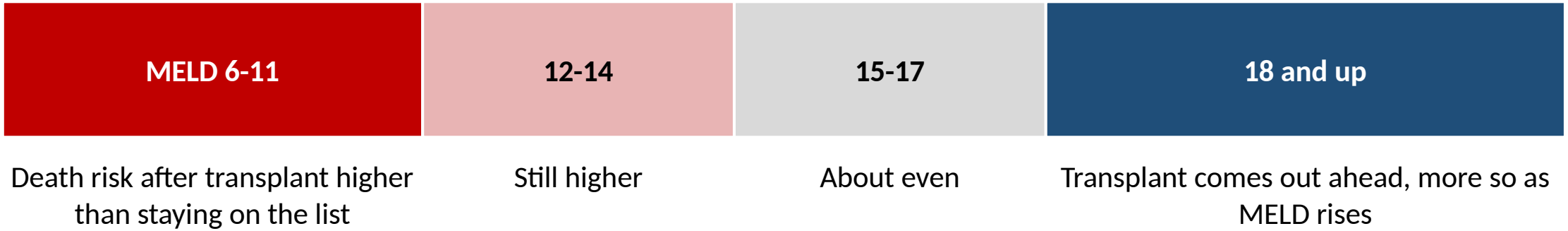
When should a patient with cirrhosis be referred?



Statement 1: refer after any decompensating event. Survival benefit may start at MELD ≥12. “MELD score alone should not be a barrier to referral for transplantation.”

Figure 1 and statement 1 redrawn from Dove L et al. AASLD/AST candidate evaluation guideline. Hepatology 2026;83(6):1609-45 [PMID: 41405234]

Why did MELD 15 become the rule?



- Survival benefit analysis of US waitlist candidates: transplant vs staying on the list, by MELD at listing
- A 2009 decision analysis came to the same threshold of 15
- Both were before MELD-Na, MELD 3.0 and today's donor pool

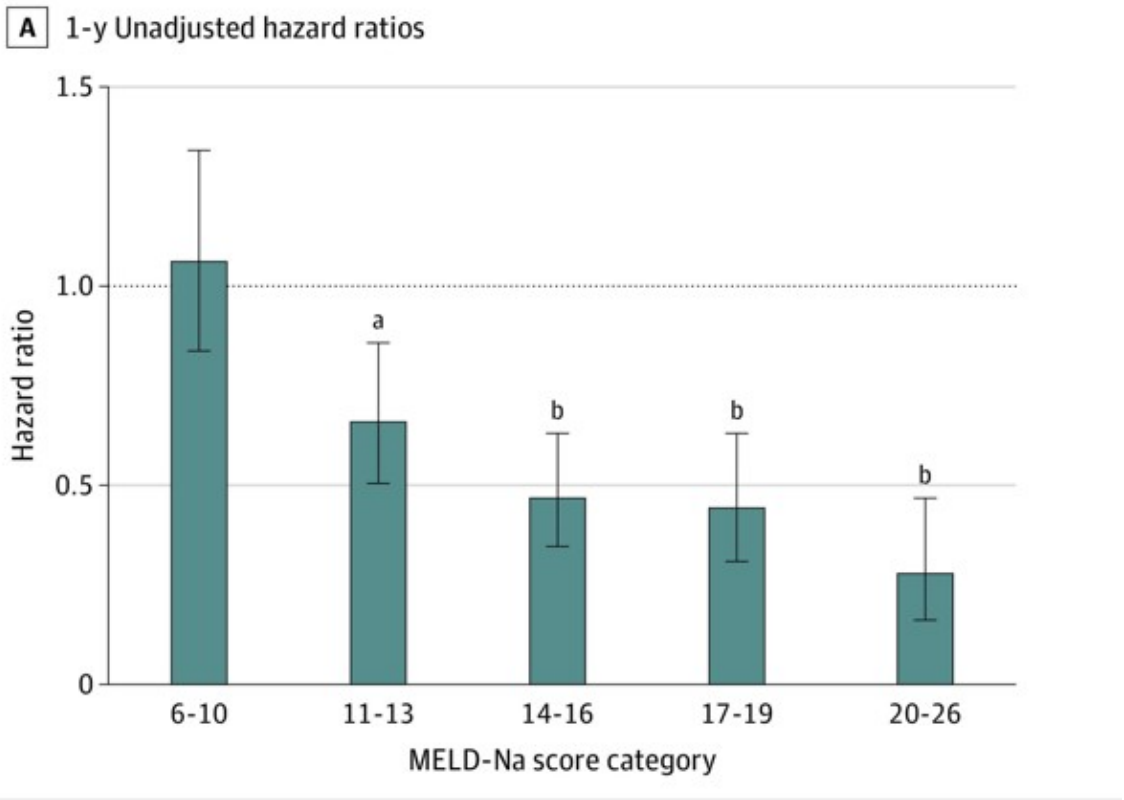
Merion RM et al. Am J Transplant 2005;5(2):307-13 [PMID: 15643990]. Liver Transpl 2009;15(2):242-54

Is there benefit below MELD 15 with a living donor?

JAMA Surg. 2022 Oct 1;157(10):926-932. doi: 10.1001/jamasurg.2022.3327.

Survival Benefit of Living–Donor Liver Transplant

Whitney E Jackson [1 2], John S Malamon [2 3], Bruce Kaplan [1 2 3], Jessica L Saben [2 3], Jesse D Schold [4], James J Pomposelli [2 3], Elizabeth A Pomfret [2 3]



Compared with staying on the list:

- MELD-Na 11-13: aHR 0.64 (0.47-0.88)
- MELD-Na 14-16: aHR 0.47 (0.34-0.66)
- Modeled gain of 13-17 life-years

Jackson WE, Malamon JS, Kaplan B, Saben JL, Schold JD, Pomposelli JJ, Pomfret EA. JAMA Surg 2022;157(10):926-932 [PMID: 35921119]

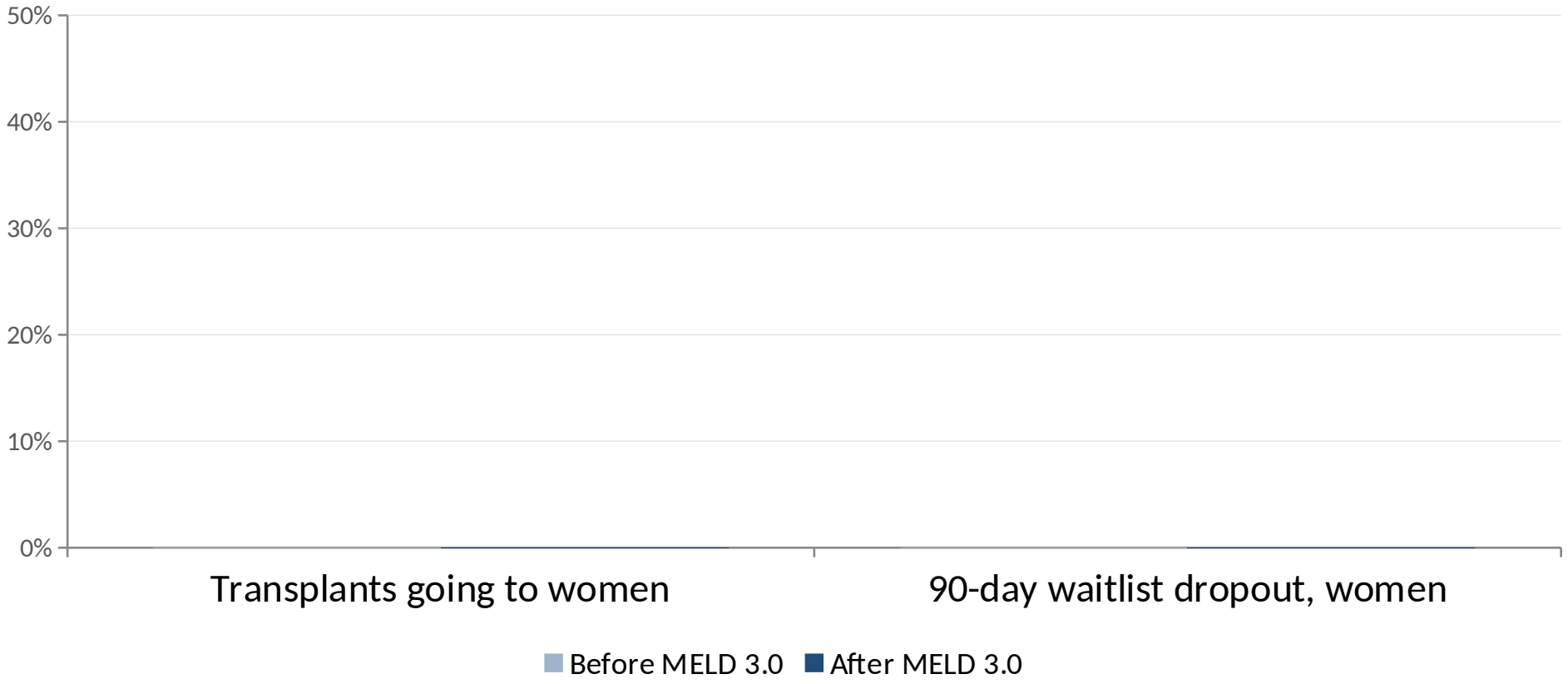
What's new in MELD 3.0?

	MELD-Na	MELD 3.0
Bilirubin	✓	✓
INR	✓	✓
Creatinine	✓ (cap 4.0)	✓ (cap 3.0)
Sodium	✓	✓
Albumin		✓
Female sex		✓ **(+1.33)**
Interaction terms	MELD × Na	Bili × Na, albumin × Cr

- Used for allocation since July 2023
- 90-day mortality: C-statistic 0.869 vs 0.862
- In simulation, at least 20 fewer waitlist deaths a year

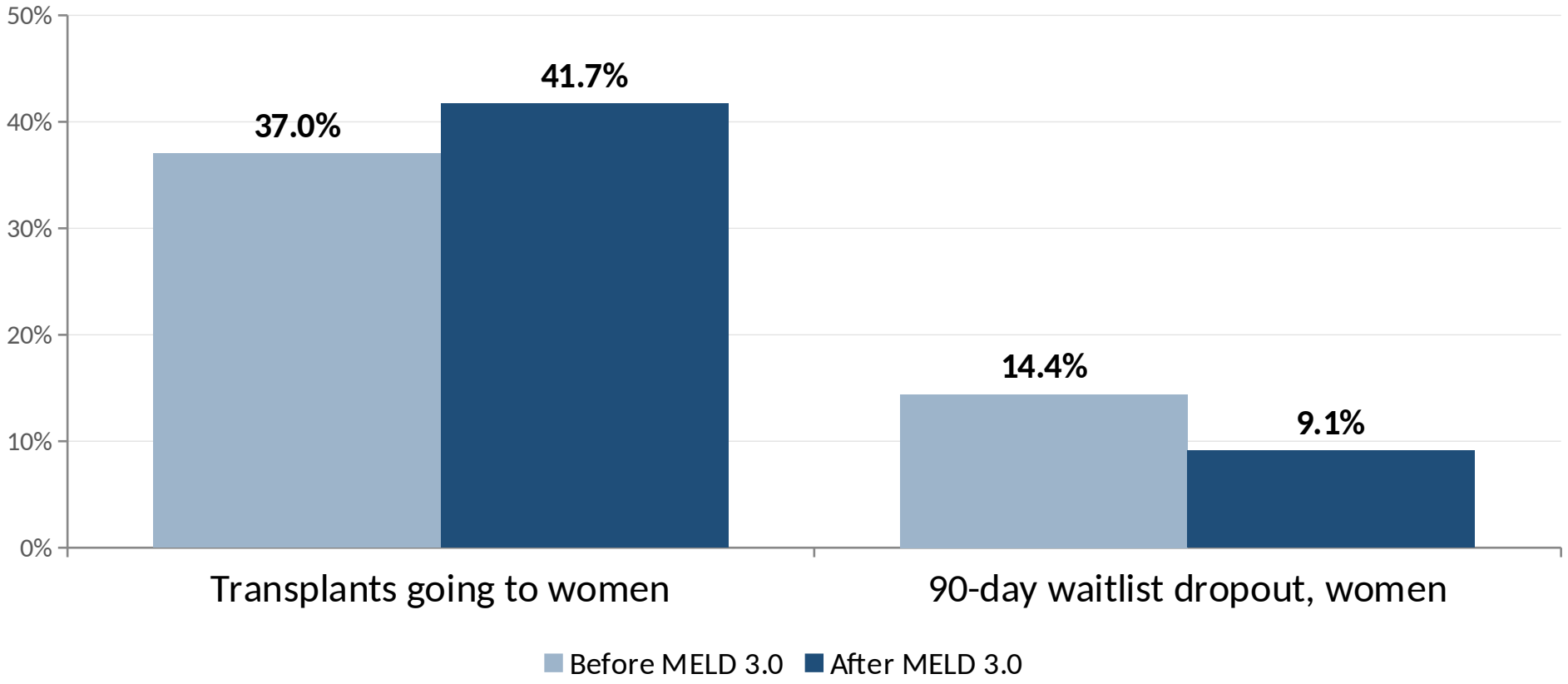
Kim WR, Mannalithara A, Heimbach JK, et al. Gastroenterology 2021;161(6):1887-1895.e4 [PMID: 34481845]

Did MELD 3.0 help women?



Kwong AJ, The Liver Meeting 2024, abstract 5002 (conference data). Tanaka T et al. Hepatology 2026;84(1):106-16 [PMID: 41100798]. Di Napoli M et al. Front Transplant 2026 [PMID: 42266471]. SRTR 2024 ADR [PMID: 42276697]

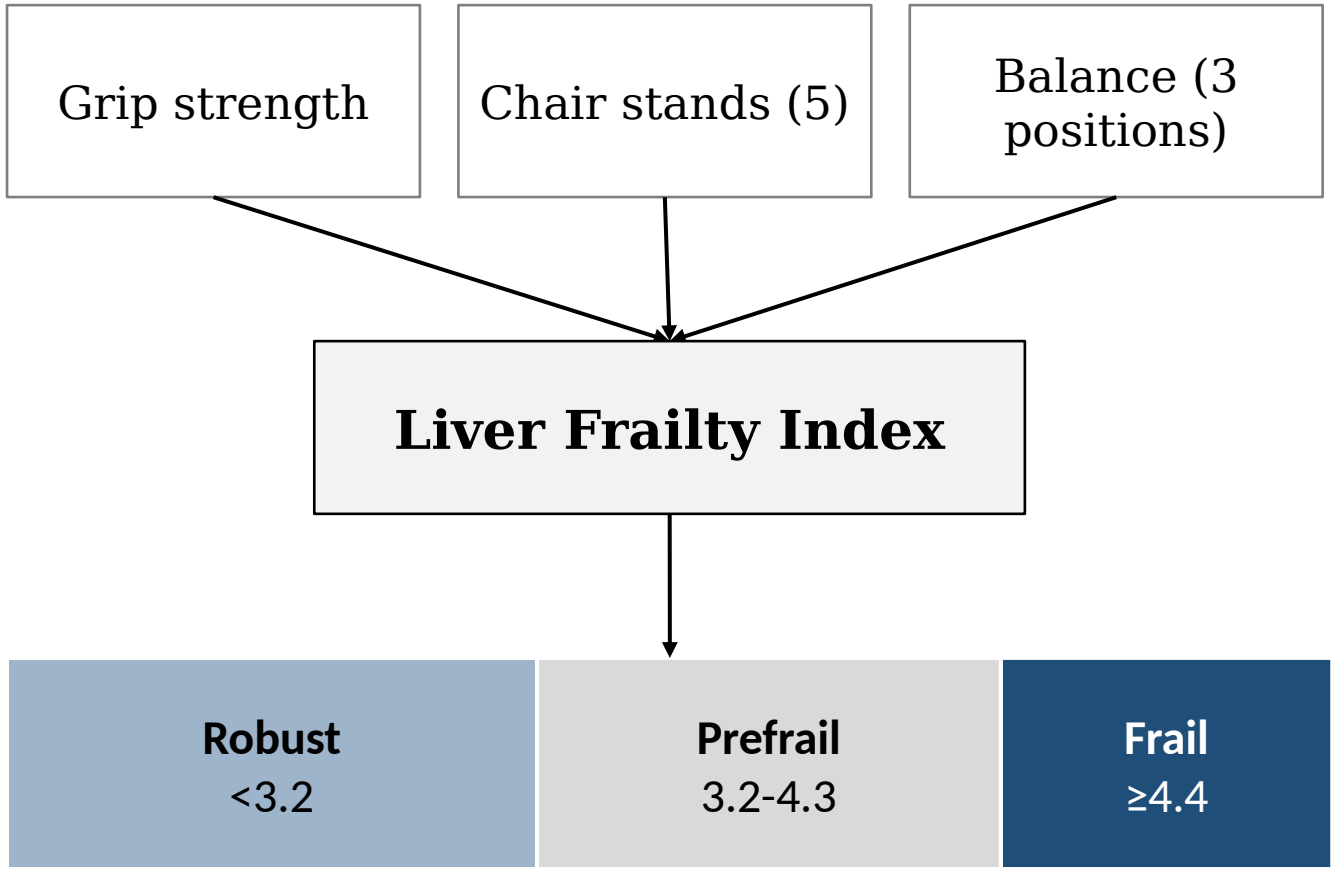
Did MELD 3.0 help women?



The gain was largest right after the change and narrowed over time. Waitlisted women still die more often: 15.0 vs 12.4 per 100 patient-years in 2024.

Kwong AJ, The Liver Meeting 2024, abstract 5002 (conference data). Tanaka T et al. Hepatology 2026;84(1):106-16 [PMID: 41100798]. Di Napoli M et al. Front Transplant 2026 [PMID: 42266471]. SRTR 2024 ADR [PMID: 42276697]

Which patients are too frail to wait?



- Measure it with a standard tool (statement 33)
- Frailty raises the risk after transplant, but alone it doesn't rule a patient out (44-45)
- Frailty also predicts death on the waitlist

Lai JC, Tandon P, Bernal W, et al. Hepatology 2021;74(3):1611-44 [PMID: 34233031]. Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234]

What no longer rules a patient out?

On its own	2025 AASLD/AST guideline	Statement
Length of abstinence	Not an exclusion criterion	37-39
Methadone or buprenorphine	No denial, no taper needed	41
Frailty	Assess it with a standard tool; not a sole contraindication	44-45
BMI >40 or <18.5	Screen nutrition and muscle; BMI alone doesn't exclude	48-49
ACLF or MELD >40	Not a contraindication	2, 43
Older age (living donor LT)	Not an exclusion	51

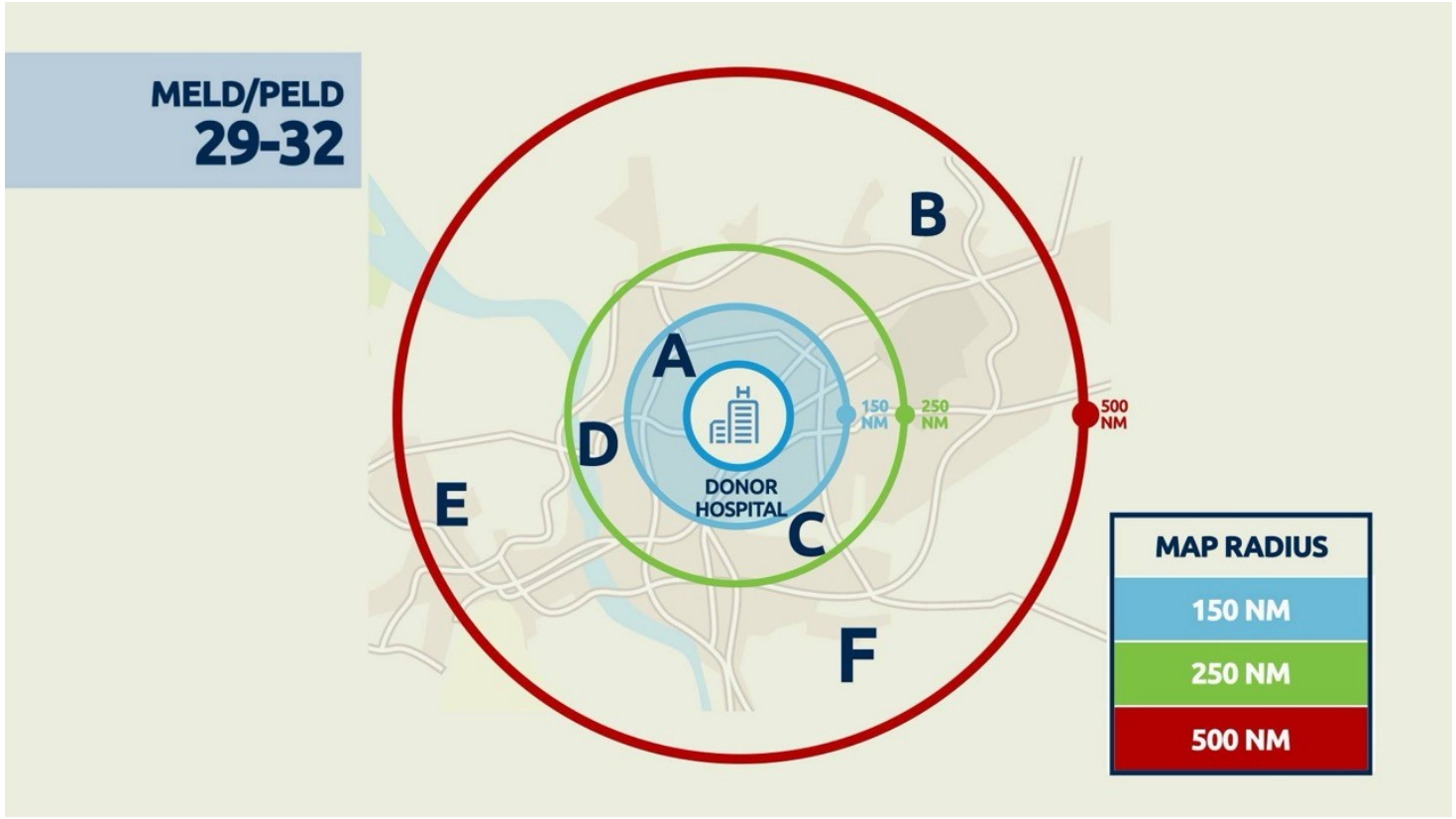
Each one is still weighed with the rest of the picture, case by case.

Dove L et al. AASLD/AST candidate evaluation guideline. Hepatology 2026;83(6):1609-45 [PMID: 41405234]

From Maine to Boston



How does a liver get from a donor to a patient?



- Circles of 150, 250 and 500 nautical miles around the donor hospital
- The sickest candidates in the closest circle get the first offers
- Exception points are set from the median MELD at transplant (MMaT) near the donor hospital

OPTN Policy 9, liver allocation (acuity circles since February 2020)

What happens after the referral?

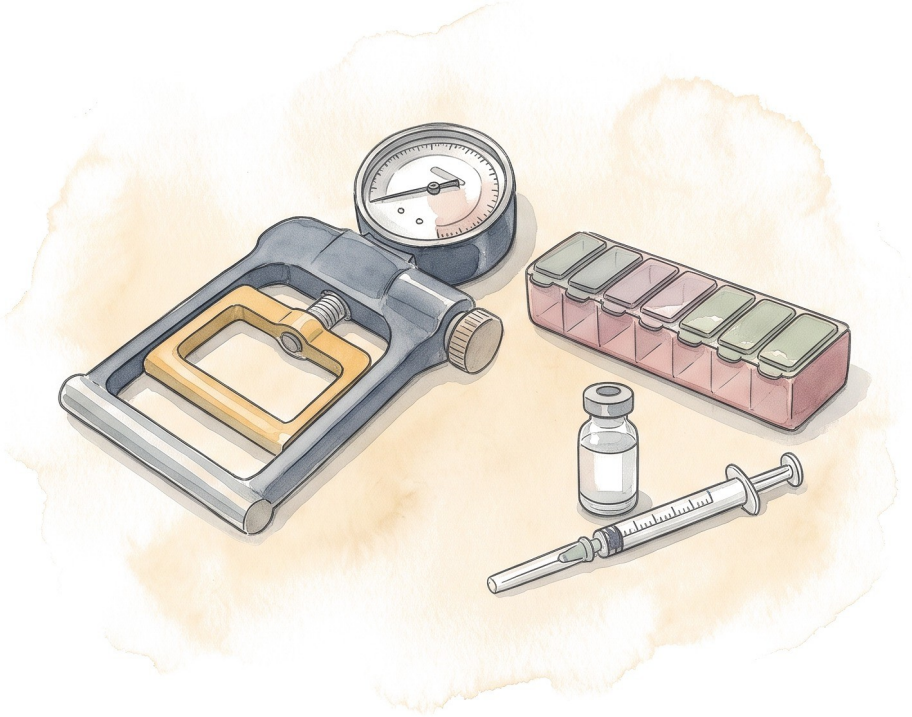


Part of the evaluation	What it involves
Hepatology and surgery	Visits, labs, CT or MRI of the liver
Heart and lungs	ECG and echo, then coronary testing by risk
Social work and addiction	Support at home, alcohol and drug history, treatment plan
Nutrition and physical therapy	Muscle, frailty, protein intake
Other	Dental, cancer screening, vaccines, financial counseling

Dove L et al. AASLD/AST candidate evaluation guideline. Hepatology 2026;83(6):1609-45 [PMID: 41405234]

What can you do while the patient waits?

- Treat alcohol use disorder, with medicines and counseling
- Nutrition: protein 1.2-1.5 g/kg a day, a late-evening snack
- Exercise for frailty
- Vaccines before transplant
- HCC surveillance every 6 months
- Keep the MELD labs current

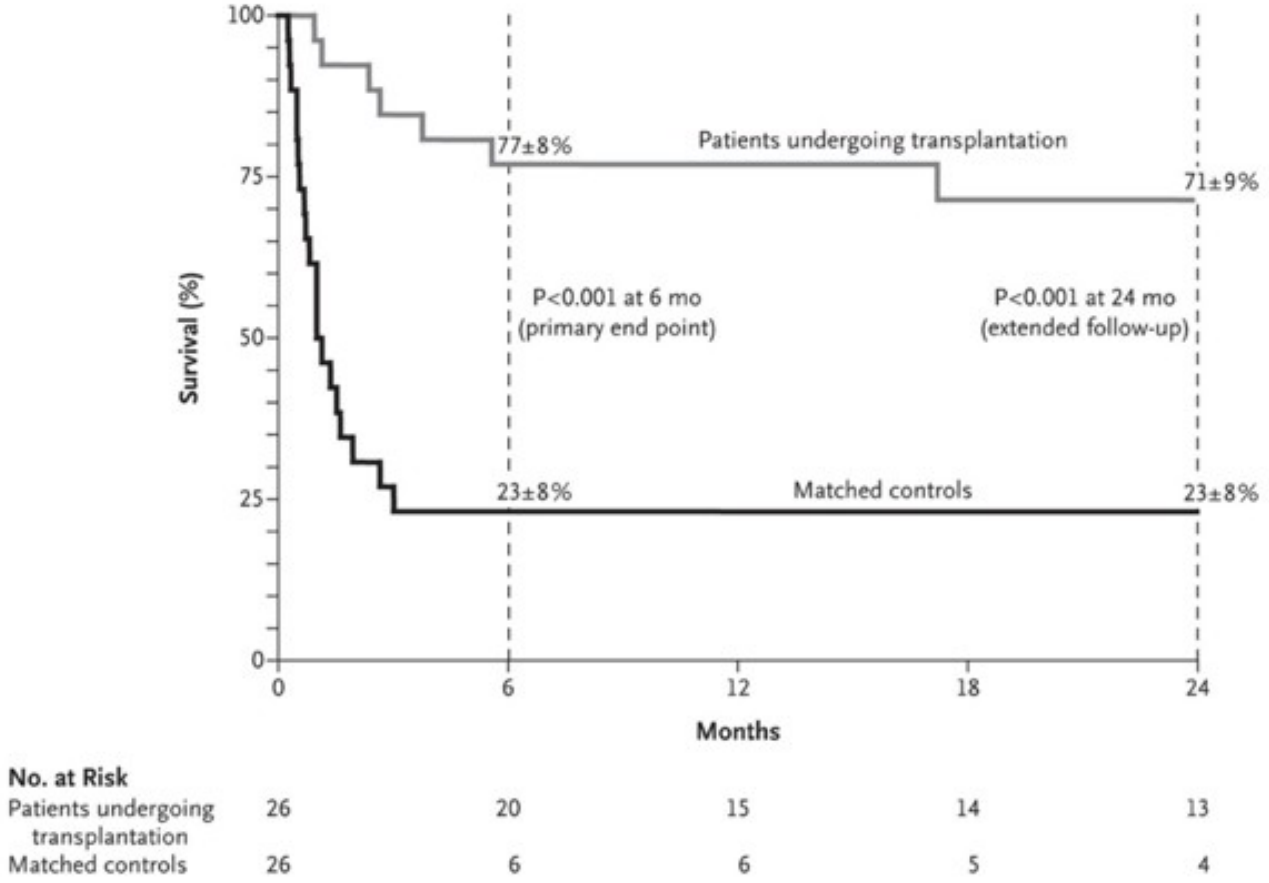


Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234]. Lai JC et al. Hepatology 2021 [PMID: 34233031]

Who can get a liver now



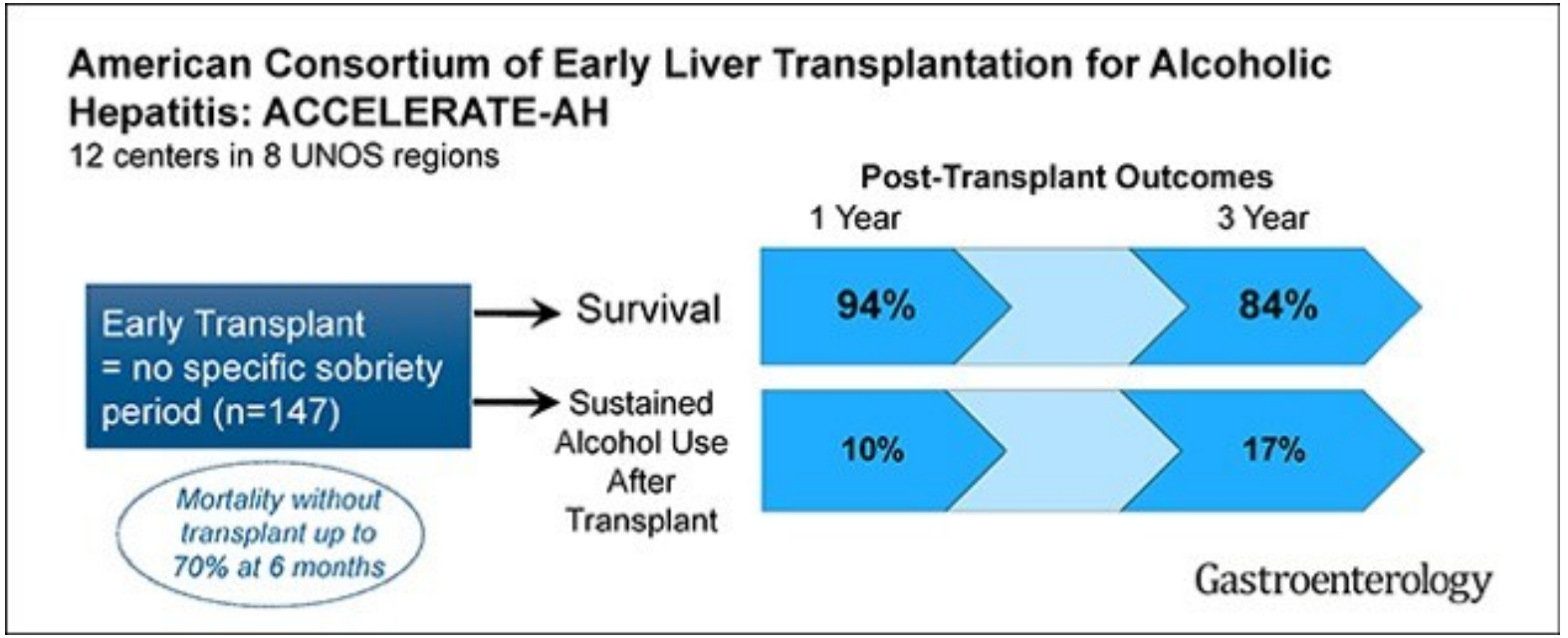
Can a patient who drank last month get a liver?



- **France and Belgium, not the US**
- Severe alcohol-associated hepatitis not responding to steroids, first liver presentation
- 26 patients, fewer than 2% of severe AH admissions
- 6-month survival 77% vs 23% in matched controls

Mathurin P, Moreno C, Samuel D, et al. N Engl J Med 2011;365(19):1790-800 [PMID: 22070476]

Did it hold up in the US?

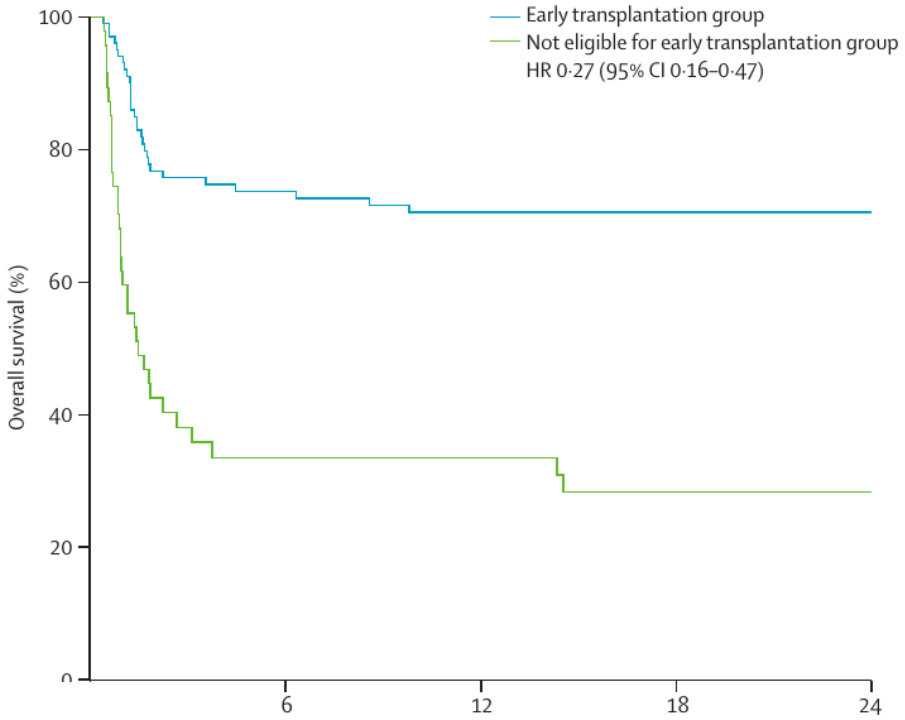


- ACCELERATE-AH, 147 patients: sustained alcohol use carried a higher risk of death (HR 4.59)
- 2025 follow-up, 347 patients: 5-year survival 83%; continued harmful drinking, aHR 3.59 for death

Lee BP et al. Gastroenterology 2018;155(2):422-430.e1 [PMID: 29655837]. Dukewich M et al. Am J Gastroenterol 2025;120(4):827-36 [PMID: 38994850]

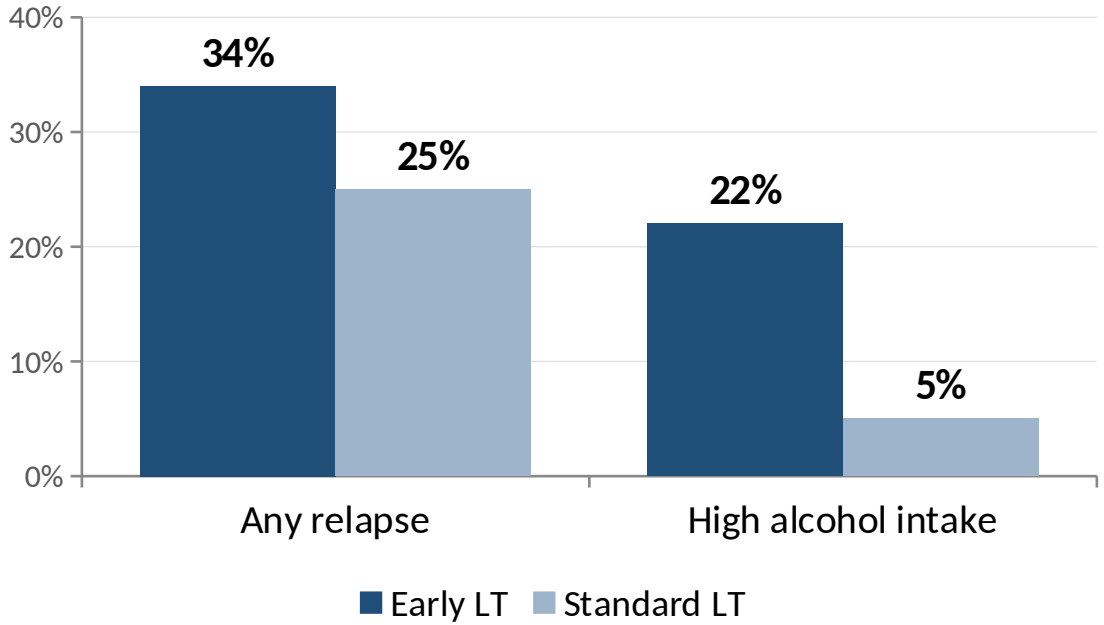
What about relapse?

Early liver transplantation for severe alcohol-related hepatitis not responding to medical treatment: a prospective controlled study



2-year survival 70.6% offered early LT vs 28.3% not eligible

Alcohol relapse after transplant



Non-inferiority for relapse not shown

Louvet A, Labreuche J, Moreno C, et al. Lancet Gastroenterol Hepatol 2022;7(5):416-25 [PMID: 35202597]

What do the guidelines say now?

Guideline	Early transplant for alcohol-associated hepatitis
EASL 2024	Selected non-responders with favorable sobriety predictors (strong)
ACG 2024	Carefully selected non-responders at low risk of relapse
AASLD/AST 2025	Refer ALD early; length of abstinence alone is not an exclusion (37-38) Methadone or buprenorphine: no denial, no taper (41)

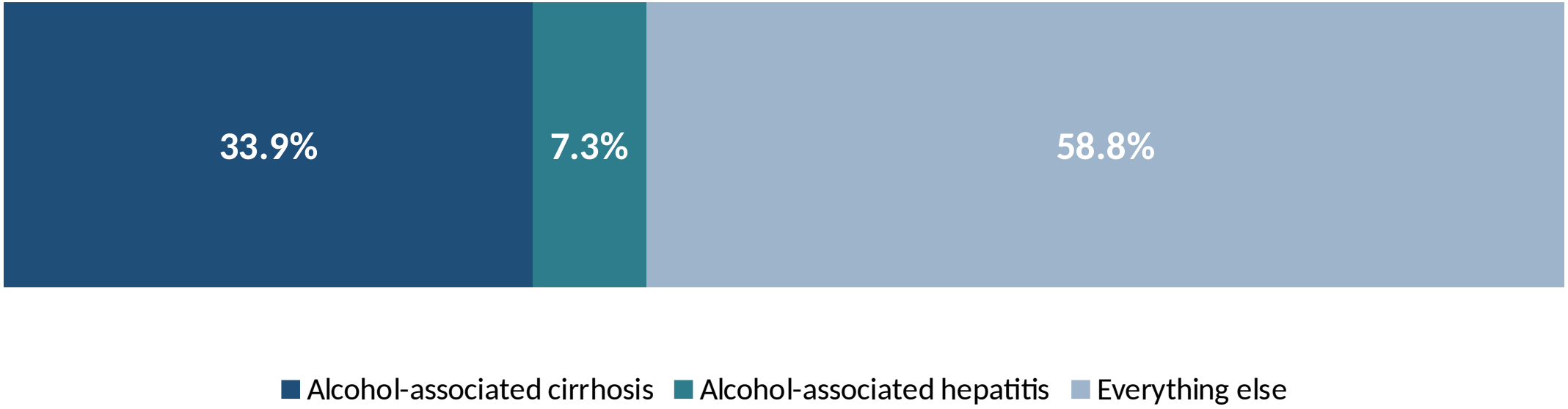
EASL CPG on liver transplantation. J Hepatol 2024;81(6):1040-86, p.1044 [PMID: 39487043]. Jophlin LL et al. Am J Gastroenterol 2024;119(1):30-54 [PMID: 38174913]. Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234]

What do US adults get a liver for?



OPTN/SRTR 2024 Annual Data Report, Table LI 7. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

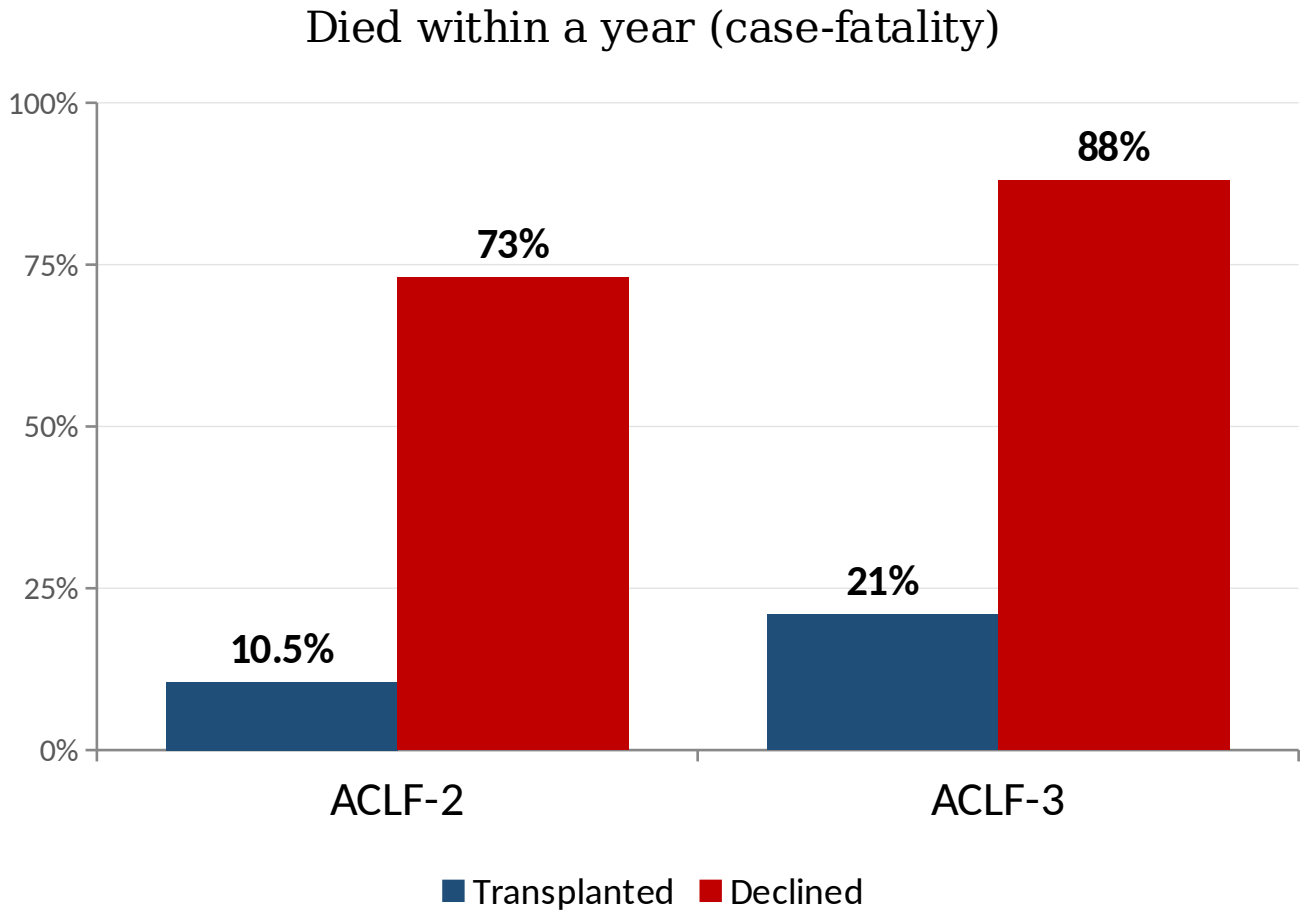
What do US adults get a liver for?



Alcohol-associated liver disease: 41.1% of US adult recipients, the top diagnosis

OPTN/SRTR 2024 Annual Data Report, Table LI 7. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

Can patients with ACLF get a liver?



- CHANCE: 612 transplanted, 144 declined, 63 centers
- No ACLF: 9% died within a year after transplant
- AASLD 2024: selected patients with ACLF may warrant expedited transplant; palliative care can run alongside the evaluation

Gustot T, Bernal W, Fernández J, et al. CHANCE. J Hepatol 2026 [PMID: 42822699]. Karvellas CJ et al. Hepatology 2024;79(6):1463-1502, statements 47, 51 [PMID: 37939273]

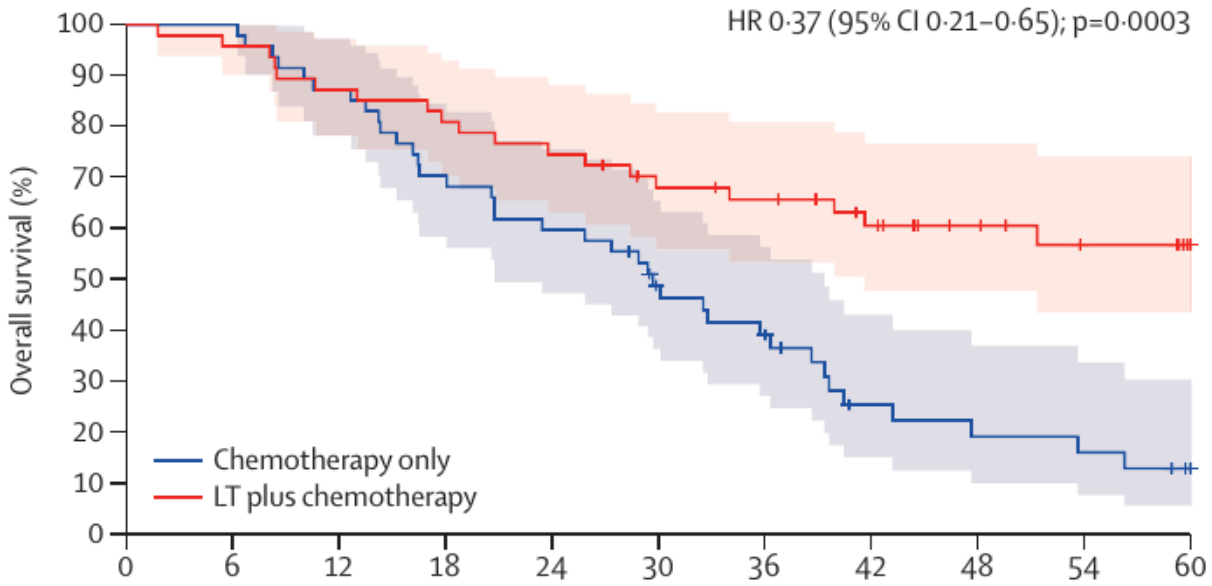
Who gets exception points?

Exception	Criteria	Score
HCC	1 lesion 2-5 cm, or 2-3 lesions each 1-3 cm AFP ≤1000 (if >1000, must fall below 500)	MELD, then MMaT-3
Hepatopulmonary syndrome	Room-air PaO2 <60 mmHg	MMaT-3
Portopulmonary HTN	Before treatment: mPAP ≥35, PVR ≥3 WU After: mPAP <35 with PVR <5, or 35-44 with PVR <3	MMaT-3
Colorectal liver metastases	Review board, since Feb 2025	MMaT-20 (floor 15)
Intrahepatic cholangiocarcinoma	Review board, since Feb 2025	MMaT-3

OPTN Policies 9.5.E, 9.5.G, 9.5.I (effective 10/1/2026); OPTN adult MELD exception transplant oncology review guidance

Can a liver transplant help colorectal metastases?

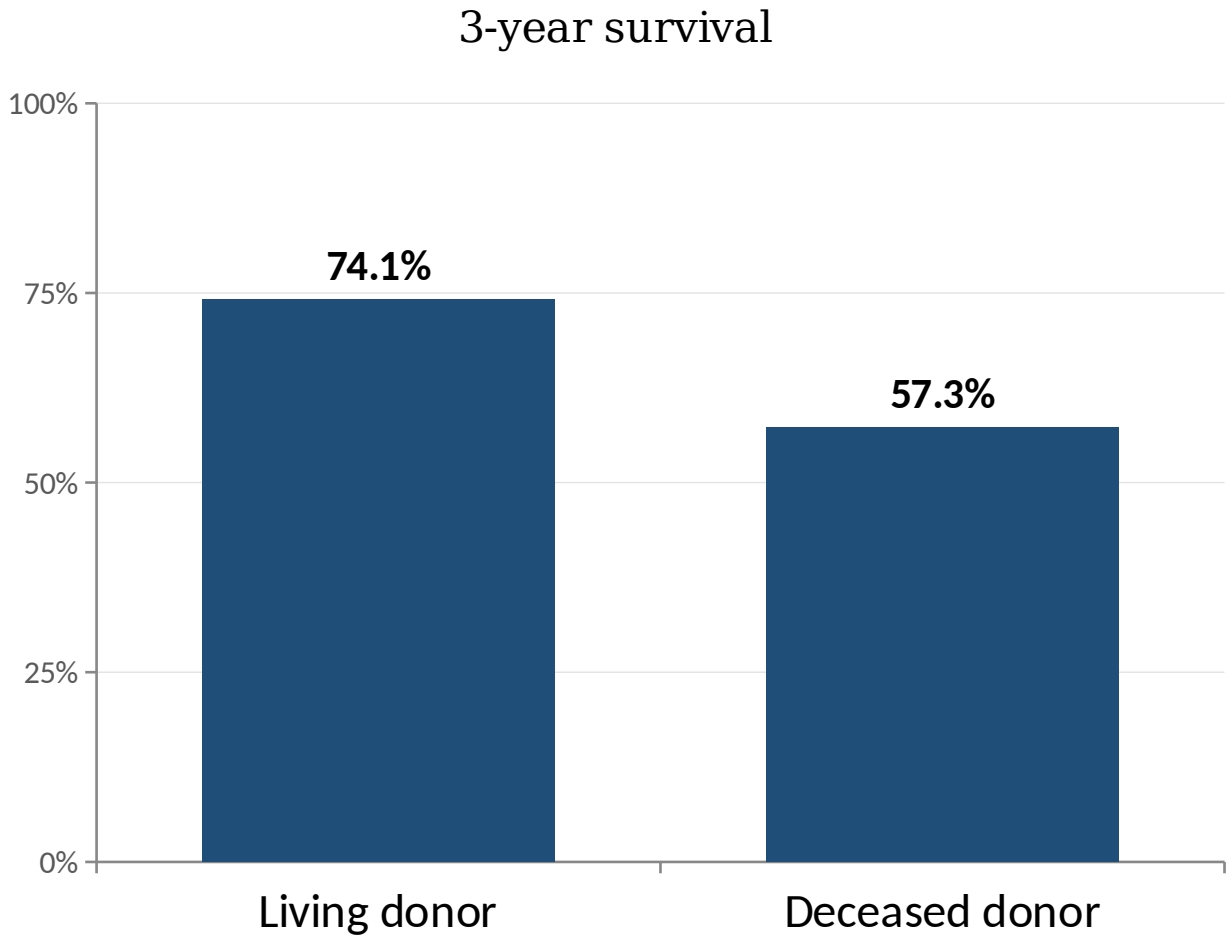
Liver transplantation plus chemotherapy versus chemotherapy alone in patients with permanently unresectable colorectal liver metastases (TransMet): results from a multicentre, open-label, prospective, randomised controlled trial



- **20 centers in France, Belgium and Italy, not the US**
- 94 randomized: age 18-65, BRAF wild-type, primary resected, liver-only disease responding to chemotherapy
- 5-year survival 56.6% vs 12.6% (HR 0.37)
- 72% of transplanted patients recurred, mostly in the lungs; almost half of recurrences were resected or ablated

Adam R, Piedvache C, Chiche L, et al. Lancet 2024;404(10458):1107-18 [PMID: 39306468]

And in the US?



- 158 transplants for colorectal metastases at 33 US centers, Dec 2017 to Mar 2025
- About half from living donors
- 3-year survival 66.7%, disease-free 44.0%
- Exception points by review since Feb 2025

Chávez-Villa M, Sasaki K, Nakayama T, et al. Liver Transpl 2026;32(4):508-17 [PMID: 41086433]

Intrahepatic cholangiocarcinoma

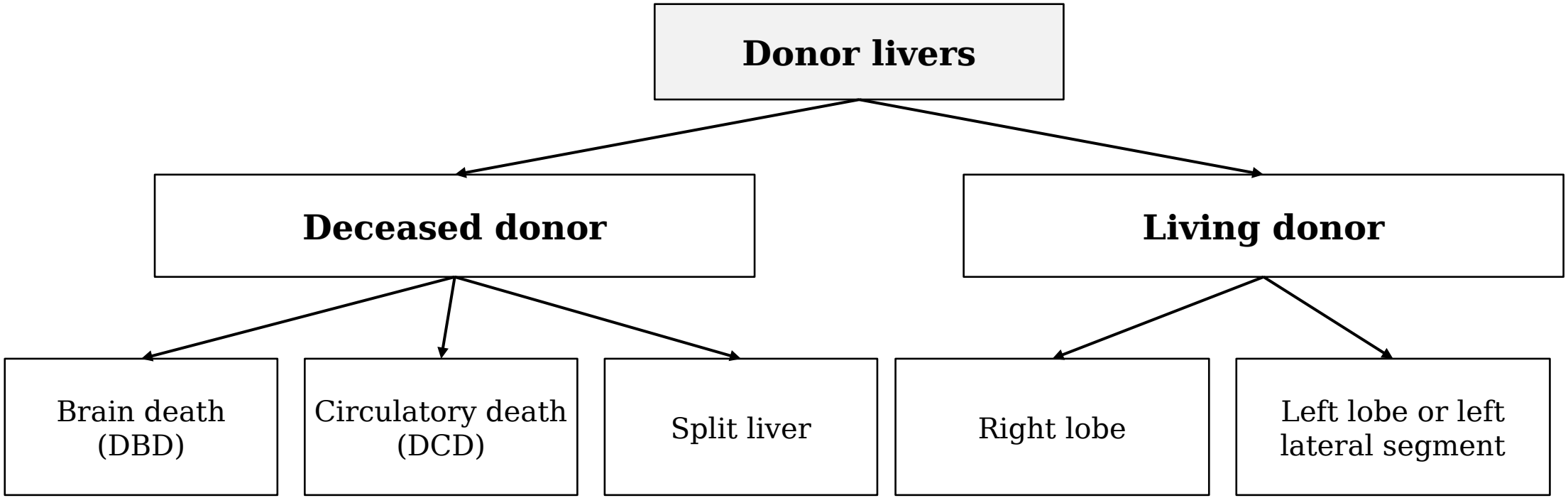
	With cirrhosis	Without cirrhosis
Tumor	≤3 cm, unresectable	Unresectable
Biopsy	Required	Required
Before listing	Stable on observation	≥6 months stable on systemic therapy
Outside the liver	No disease	No disease
Setting	Exception points by review (MMaT-3)	Research protocols only

Kodali S, Kulik L, D'Allessio A, et al. ILTS-ILCA consensus. Liver Transpl 2025;31(6):815-31 [PMID: 40014003]. Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234], statement 11

Where the livers come from



What kinds of donors are there?



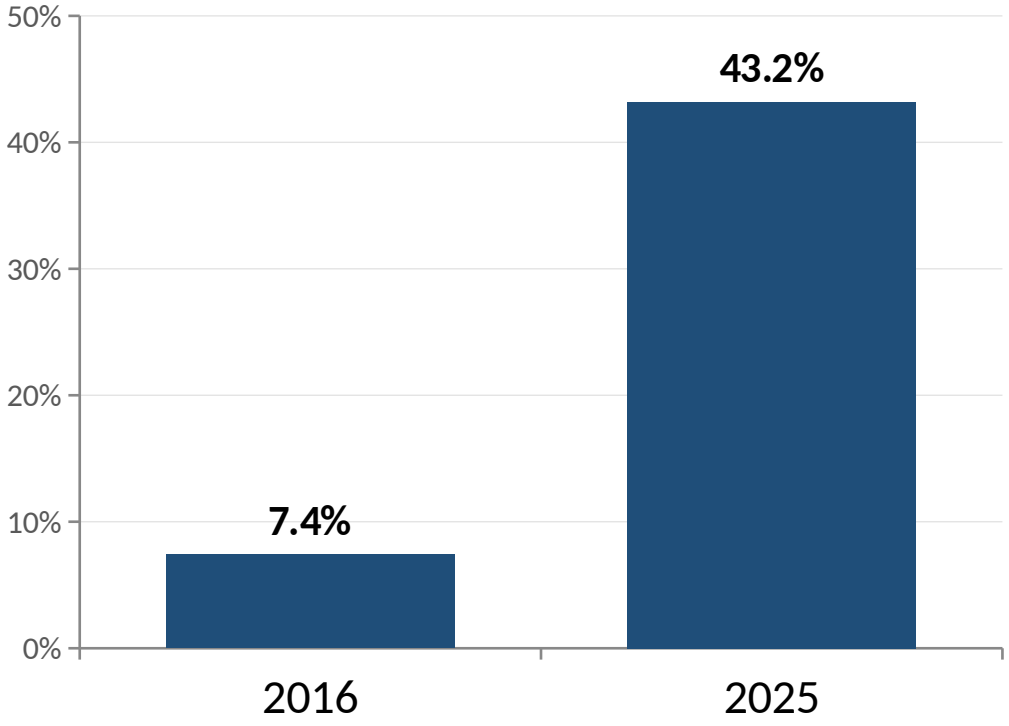
DCD: 43% of US deceased donor liver transplants in 2025

Living donor: 4.5% of US adult liver transplants in 2024

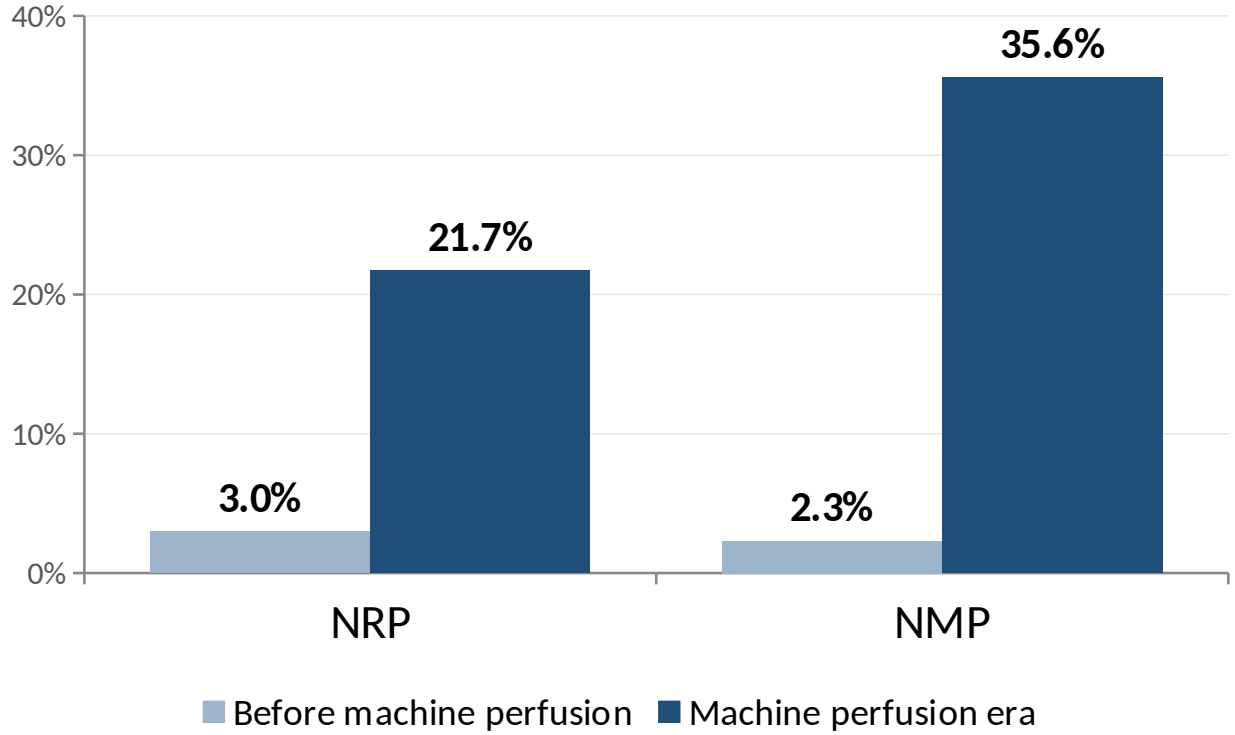
Giorgakis E et al. Transpl Int 2026;39:16689 [PMID: 42294094]. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

How much has DCD grown?

DCD share of deceased donor LTs



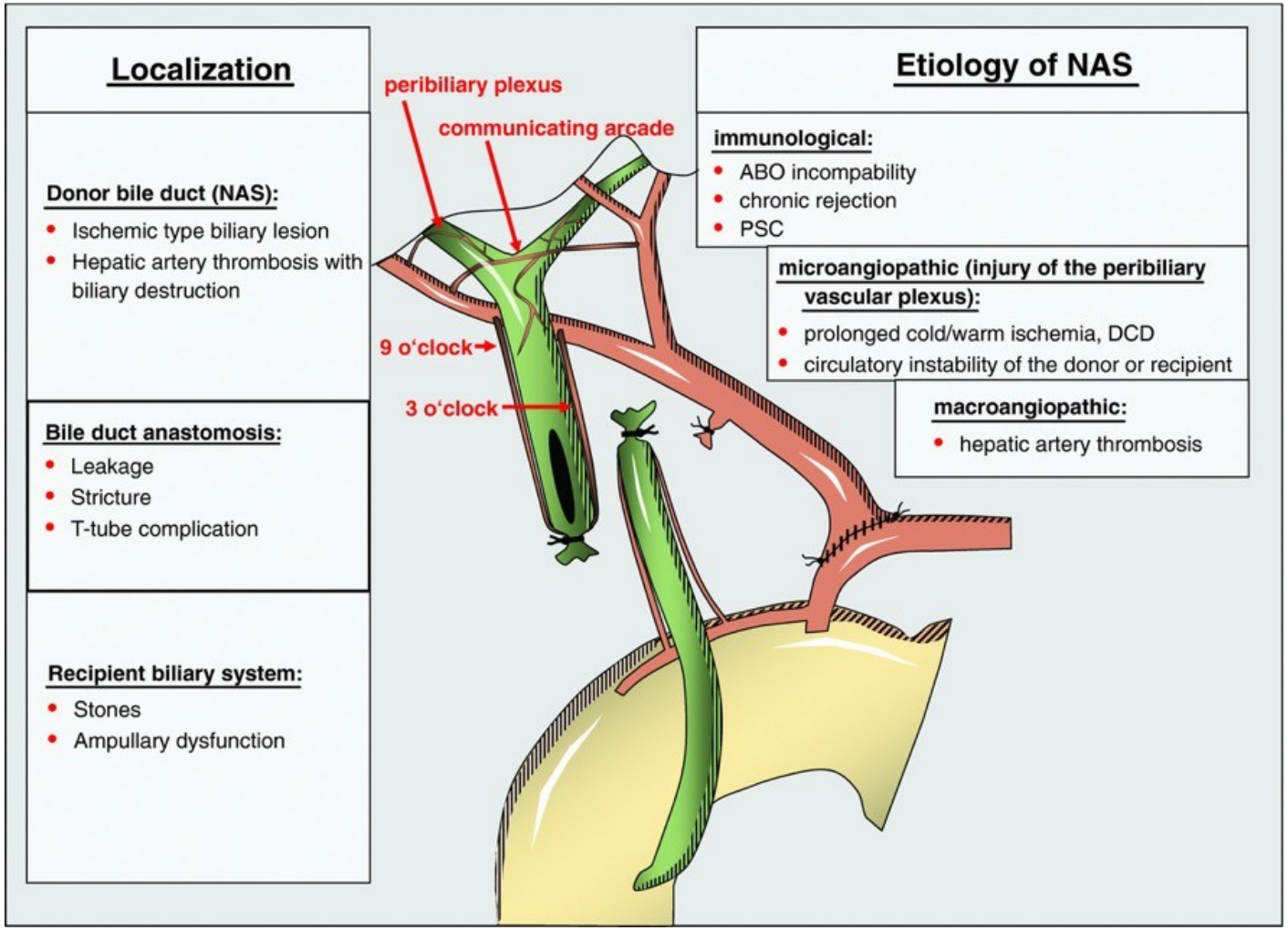
Use of NRP and NMP



Median time on the waitlist fell from 112 to 62 days

Giorgakis E, Ziogas IA, Moris D, et al. Transpl Int 2026;39:16689 [PMID: 42294094]

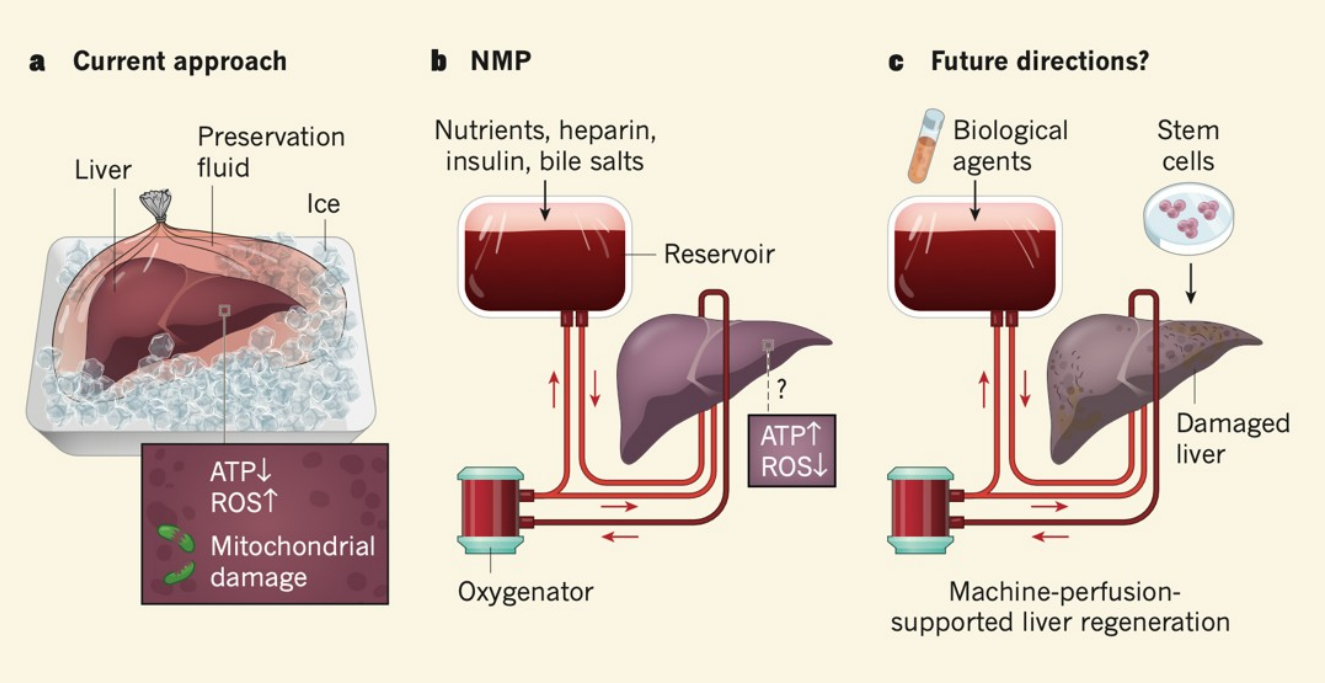
Why did DCD livers worry us?

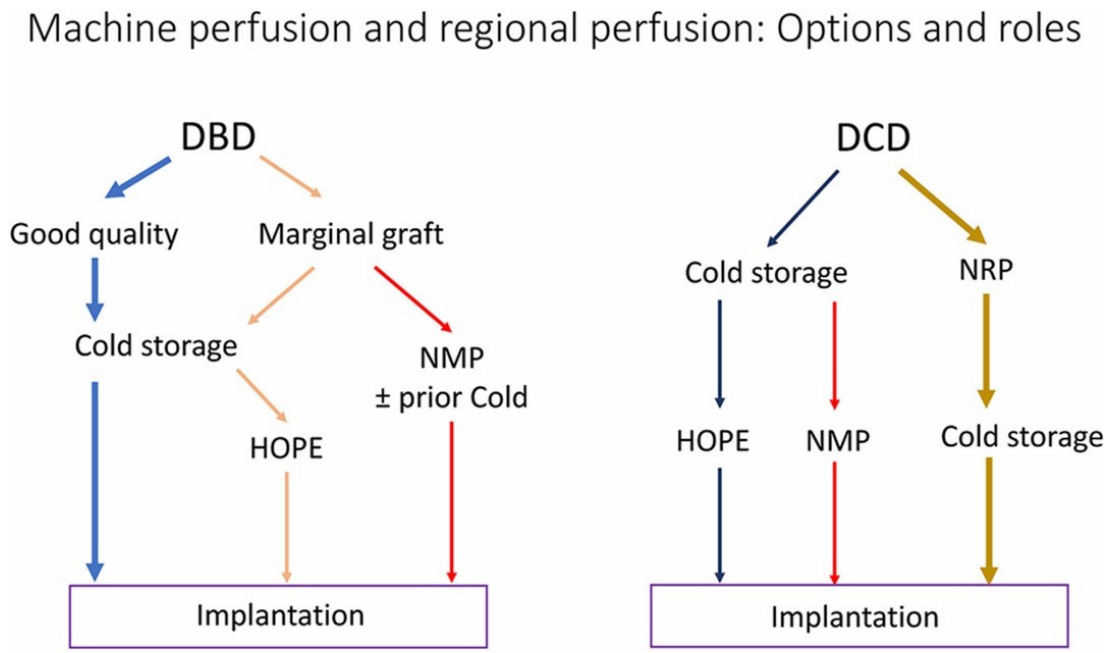


- The bile ducts get their blood only from the hepatic artery, through the peribiliary plexus
- Warm ischemia during DCD injures that plexus and the duct lining
- Weeks to months later: non-anastomotic strictures, sometimes needing retransplant

Seehofer D, Eurich D, Veltzke-Schlieker W, Neuhaus P. Am J Transplant 2013;13(2):253-65

What does a perfusion machine do?



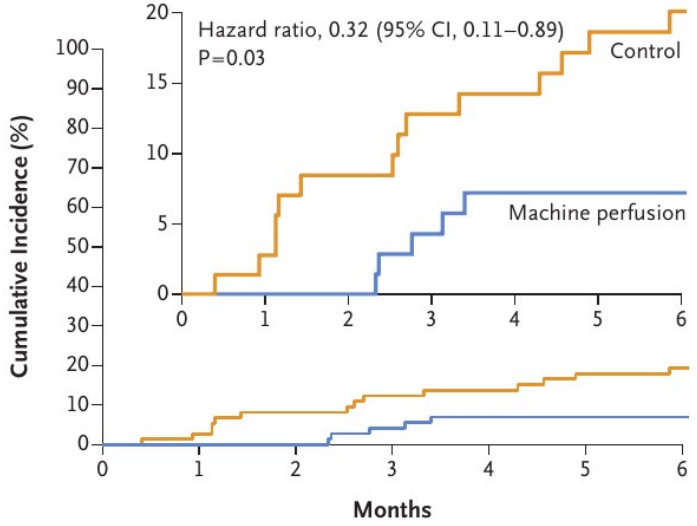


- **HOPE:** cold and oxygenated, for 1-2 hours at the transplant center before implant
- **NMP:** warm and blood-based; the liver makes bile and can be assessed before use
- **NRP:** the donor's own abdominal organs are reperfused after circulatory death

Schneeberger S. Nature 2018;557(7703):40-1. J Clin Exp Hepatol 2024;14(2):101309

What did the trials show?

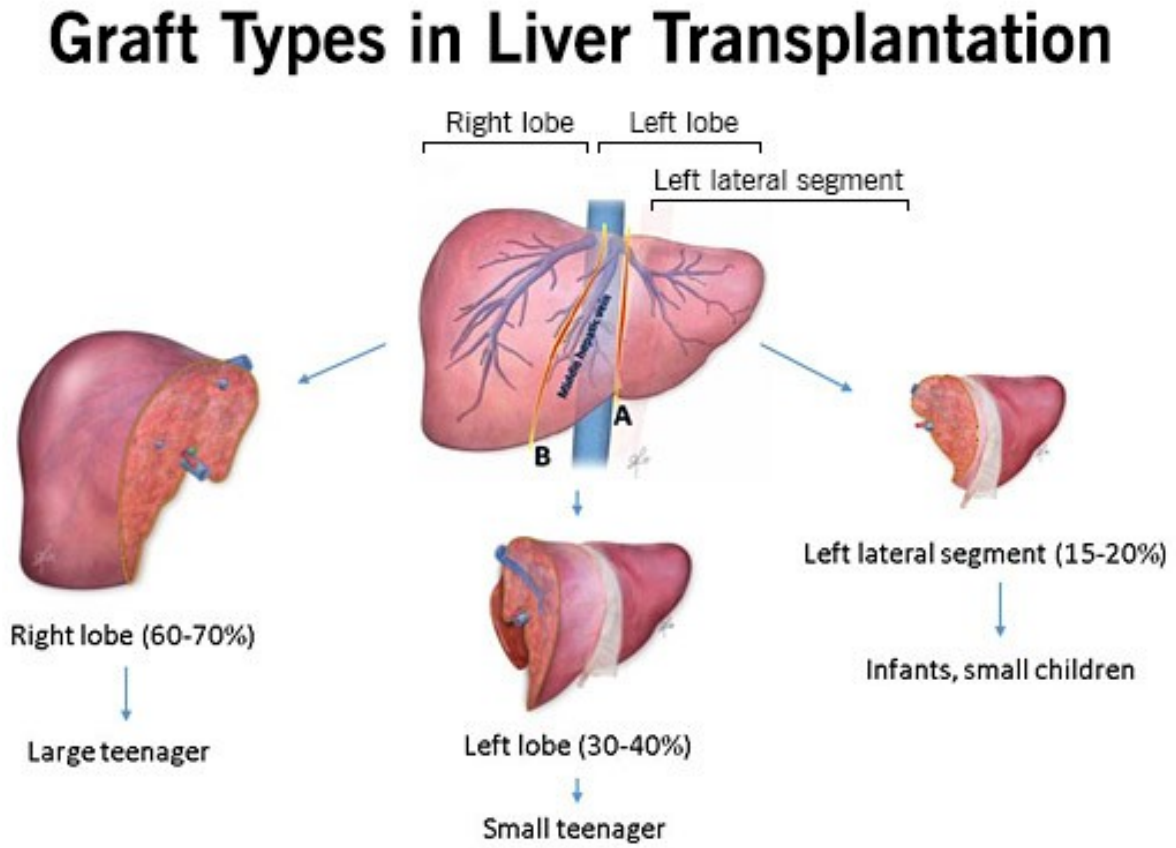
Trial	Machine	Main result
Nasralla, Nature 2018 (220 LTs)	NMP	Graft injury and discard 50% lower
PROTECT, JAMA Surg 2022	NMP (OCS)	EAD 18% vs 31% Biliary 2.6% vs 9.9% at 1 year
DHOPE-DCD, NEJM 2021	HOPE	Symptomatic NAS 6% vs 18% at 6 months (RR 0.36)



The 2025 AASLD/AST and 2024 EASL guidelines both recommend machine perfusion for DCD and extended-criteria livers.

Nasralla D et al. Nature 2018 [PMID: 29670285]. Markmann JF et al. JAMA Surg 2022 [PMID: 34985503]. van Rijn R et al. N Engl J Med 2021 [PMID: 33626248]. Te HS et al. 2026, statements 6-10 [PMID: 40844852]. EASL 2024, p.1062

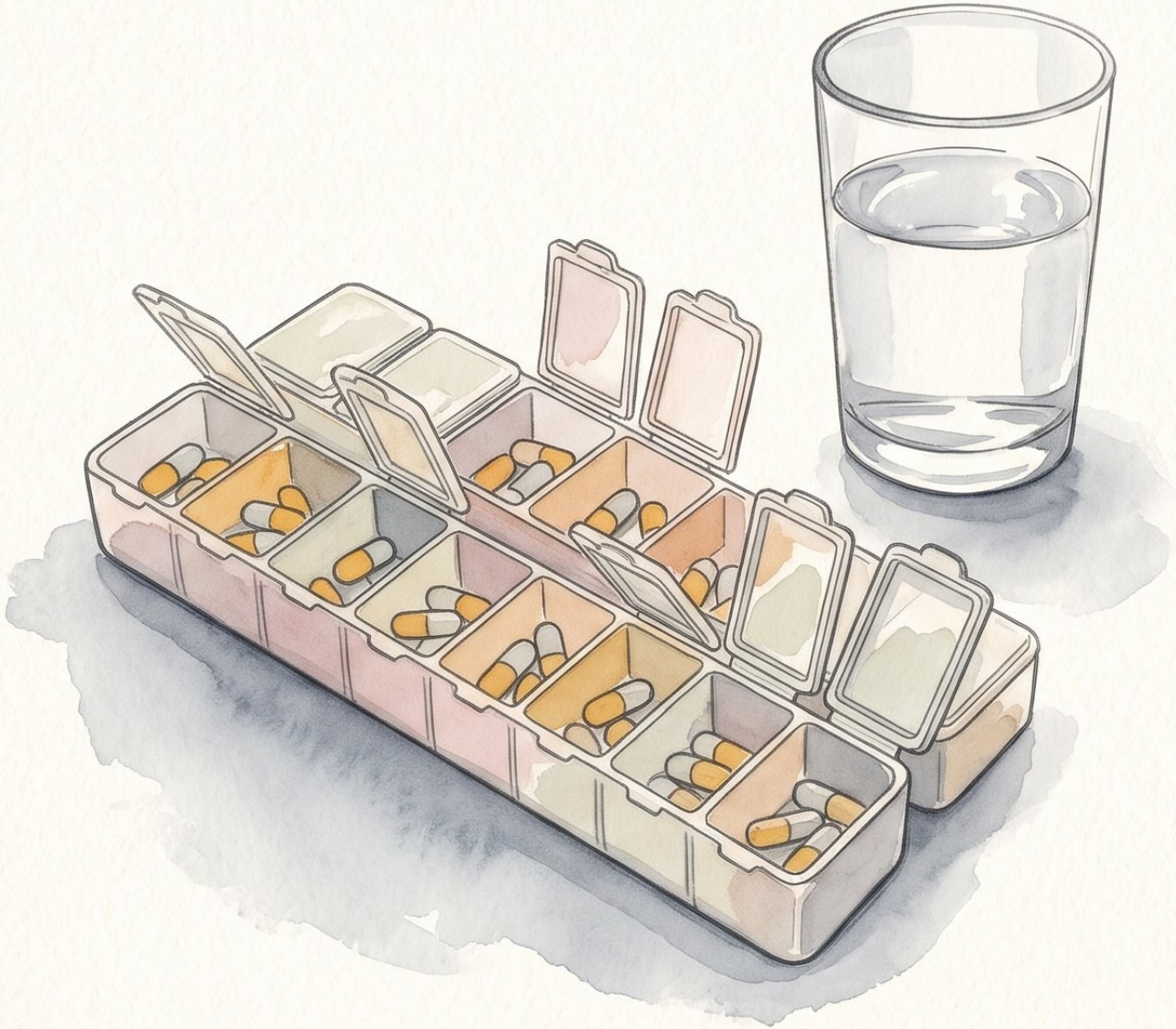
What about living donors?



- 4.5% of US adult liver transplants in 2024
- Graft-to-recipient weight ratio at least 0.8%
- Caution above MELD 30
- Older recipient age alone doesn't exclude
- The benefit starts at low MELD (Jackson 2022)

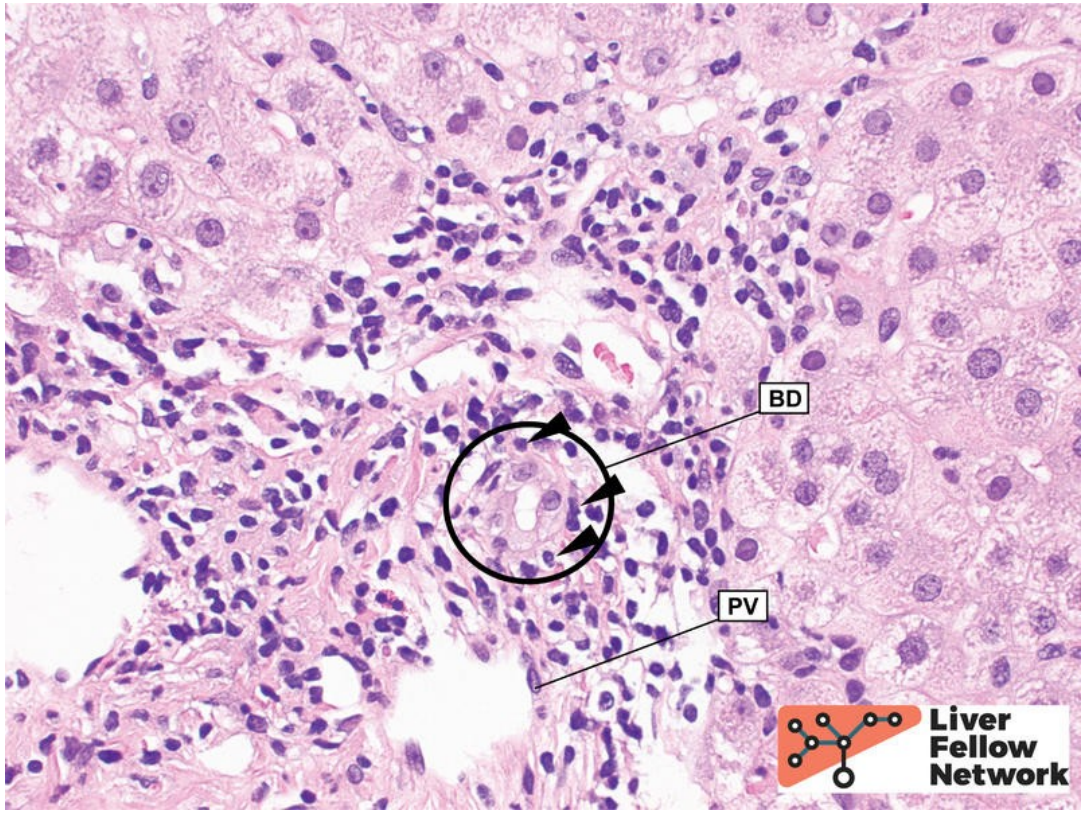
Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234], statements 50-53. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

Your patient had a transplant



Abnormal liver tests after transplant: what next?

- Treat it as rejection until proven otherwise
- Same day: Doppler ultrasound and a tacrolimus level
- Call transplant hepatology
- Biopsy is the gold standard; the transplant team decides on it



Acute cellular rejection

Te HS et al. AASLD/AST graft-related complications guideline. Liver Transpl 2026;32(3):444-90 [PMID: 40844852], statement 34. Histology: Liver Fellow Network

MRCP or ERCP for a suspected stricture?



Situation	Next step
Suspected stricture	**MRCP first (strong)**
High suspicion, or intervention likely	ERCP directly (weak)
Normal MRCP, picture still fits	ERCP (weak)
Irreversible ischemic cholangiopathy	Retransplant

Te HS et al. AASLD/AST graft-related complications guideline. Liver Transpl 2026;32(3):444-90 [PMID: 40844852], statements 15, 16, 24. Kohli DR et al. ASGE guideline. Gastrointest Endosc 2023;97(4):607-14

Which drugs change tacrolimus levels?

Raise levels (toxicity)	Lower levels (rejection)
Fluconazole, voriconazole, posaconazole	Rifampin
Clarithromycin, erythromycin	Phenytoin, carbamazepine
Diltiazem, verapamil	Phenobarbital
Nirmatrelvir-ritonavir, other boosted HIV drugs	St John's wort
Grapefruit	

No herbal supplements without asking the transplant team, and no grapefruit (statements 2-3).

Tacrolimus (Prograf) prescribing information. Sharma P et al. AASLD/AST non-graft complications guideline. Liver Transpl 2026;32(8):1157-93 [PMID: 41405235]

What does the primary care doctor need to do?

	2025 guideline	Statement
Screening	BP, weight, glucose and lipids every year	53
Type 2 diabetes	Metformin first; GLP-1 RA or SGLT2i if high CV or kidney risk	58-59
Hypertension	Amlodipine or felodipine; ACEi or ARB if CVD or CKD	60-61
Lipids	Hydrophilic statin first	62
Obesity	GLP-1 RA	64
CKD	Nephrology at stage 4; the team lowers tacrolimus, don't stop it	76, 79, 82

Sharma P et al. AASLD/AST non-graft complications guideline. Liver Transpl 2026;32(8):1157-93 [PMID: 41405235]

What else does the primary care doctor need to do?

	2025 guideline	Statement
Cancer	Age-appropriate screening; skin exam every year	44-45
Vaccines	ACIP schedule for immunocompromised adults; no live vaccines	19-20
Alcohol	None, whatever the original liver disease	9
Elective surgery	At least 3 months after LT; the team manages immunosuppression	83-84
Pregnancy	Wait at least a year; stop mycophenolate beforehand	24, 26

Sharma P et al. AASLD/AST non-graft complications guideline. Liver Transpl 2026;32(8):1157-93 [PMID: 41405235]

Take-home points

- Refer after any decompensating event, or from MELD 12. A low MELD alone is not a reason to wait
- Recent drinking alone is not a reason to say no. Call transplant hepatology
- Colorectal metastases and selected cholangiocarcinoma are worth a call
- After transplant, abnormal liver tests mean a call to transplant hepatology the same day
- Check every new drug against tacrolimus



Questions?



Appendix: sources (1)

- AASLD/AST candidate evaluation guideline. Hepatology 2026 [PMID: 41405234]
- AASLD/AST graft-related complications guideline. Liver Transpl 2026 [PMID: 40844852]
- AASLD/AST non-graft complications guideline. Liver Transpl 2026 [PMID: 41405235]
- EASL CPG on liver transplantation. J Hepatol 2024 [PMID: 39487043]
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- Lai JC et al. Frailty and sarcopenia guidance. Hepatology 2021 [PMID: 34233031]
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- OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026 [PMID: 42276697]
- OPTN Policy 9 (effective 10/1/2026) and transplant oncology review guidance
- Kim WR et al. MELD 3.0. Gastroenterology 2021 [PMID: 34481845]
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- Di Napoli M et al. Front Transplant 2026 [PMID: 42266471]
- Merion RM et al. Survival benefit of LT. Am J Transplant 2005 [PMID: 15643990]

Appendix: sources (2)

- Jackson WE et al. Living donor LT benefit. JAMA Surg 2022 [PMID: 35921119]
- Mathurin P et al. Early LT for alcoholic hepatitis. N Engl J Med 2011 [PMID: 22070476]
- Lee BP et al. ACCELERATE-AH. Gastroenterology 2018 [PMID: 29655837]
- Dukewich M et al. Early LT for AH, 5-year outcomes. Am J Gastroenterol 2025 [PMID: 38994850]
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- Nasralla D et al. Nature 2018 [PMID: 29670285]; Markmann JF et al. PROTECT. JAMA Surg 2022 [PMID: 34985503]
- van Rijn R et al. DHOPE-DCD. N Engl J Med 2021 [PMID: 33626248]
- Seehofer D et al. Am J Transplant 2013; Kohli DR et al. ASGE. Gastrointest Endosc 2023